

BUILD SPECIFICATION / PART 2 OF 2 // v1.0 // SEPTEMBER 2026

Low-Level Design

Agency Portfolio Website

Design system, components, pages, schemas, API, testing and deployment runbook

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1. Agent Brief and Conventions

This Low-Level Design gives exact implementation detail. Read the HLD for context. Follow these conventions throughout:

- Language: JavaScript (ES2022) with ES modules on both client and server. JSX for React. No TypeScript in v1 (keep types via Zod schemas and JSDoc where helpful).
- Naming: components `PascalCase.jsx`, hooks `useThing.js`, utilities `camelCase.js`, server files `thing.type.js` (for example `project.routes.js`). One component per file, default export.
- Formatting: Prettier (single quotes, semicolons, 100 columns, trailing commas). ESLint with react, react-hooks and jsx-a11y plugins on the client; eslint recommended on the server.
- Commits: conventional commits (`feat:`, `fix:`, `chore:`). One commit per finished task inside a phase.
- All environment access goes through `server/src/config/env.js` (validated with Zod) and `import.meta.env` on the client. Never read `process.env` elsewhere.
- Every list, grid and form must implement loading, empty, error and success states.

2. Repository Structure and Tooling

```
agency-site/
├── client/
│   ├── public/
│   │   ├── favicon.svg
│   │   ├── og-image.png
│   │   ├── robots.txt
│   │   └── placeholders/*.svg
│   ├── scripts/generate-sitemap.js
│   │   └── runs before build, fail-soft
│   └── src/
│       ├── main.jsx
│       ├── App.jsx
│       ├── routes.jsx
│       ├── styles/index.css
│       ├── config/site.js
│       ├── content/
│       │   ├── tokens
│       │   ├── base layer
│       │   ├── utilities
│       │   ├── brand name
│       │   ├── nav links
│       │   ├── socials
│       │   ├── contact details
│       │   ├── services
│       │   ├── pricing
│       │   ├── faq
│       │   ├── team
│       │   ├── values
│       │   ├── process
│       │   ├── testimonials
│       │   ├── techStack
│       │   └── stats (all .js)
│       ├── lib/
│       │   ├── api.js
│       │   ├── queryClient.js
│       │   ├── utils.js
│       │   ├── motion.js
│       │   ├── seo.js
│       │   ├── useProjects
│       │   ├── useProject
│       │   ├── useSubmitInquiry
│       │   ├── useSpotlight
│       │   ├── useCountUp
│       │   └── useMediaQuery
│       ├── context/AuthContext.jsx
│       ├── components/ui/
│       │   ├── Button
│       │   ├── Card
│       │   ├── Badge
│       │   ├── Container
│       │   ├── Section
│       │   └── SectionHeader
│       ├── components/layout/
│       │   ├── Navbar
│       │   ├── Footer
│       │   ├── PageTransition
│       │   ├── Aurora
│       │   ├── NoiseOverlay
│       │   └── ScrollToTop
│       ├── components/sections/
│       │   ├── home/
│       │   ├── about/
│       │   ├── pricing/
│       │   ├── contact/
│       │   └── projects/
│       ├── pages/
│       │   ├── Home
│       │   ├── About
│       │   ├── Pricing
│       │   ├── Contact
│       │   ├── ProjectDetail
│       │   ├── NotFound
│       │   ├── StyleGuide
│       │   ├── Login
│       │   ├── Dashboard
│       │   ├── Projects
│       │   ├── ProjectForm
│       │   ├── Inquiries
│       │   └── AdminLayout
│       ├── index.html
│       ├── vite.config.js
│       ├── tailwind.config.js
│       ├── postcss.config.js
│       └── vercel.json
├── server/
│   └── src/
│       ├── server.js
│       ├── app.js
│       ├── config/
│       │   ├── env.js
│       │   ├── db.js
│       │   ├── cloudinary.js
│       │   └── mailer.js
│       ├── models/
│       │   ├── Project.js
│       │   ├── Inquiry.js
│       │   └── User.js
│       ├── routes/
│       │   ├── index.js
│       │   ├── auth.routes.js
│       │   ├── project.routes.js
│       │   ├── inquiry.routes.js
│       │   └── admin.routes.js
│       ├── controllers/
│       │   ├── auth
│       │   ├── project
│       │   ├── inquiry
│       │   ├── admin
│       │   └── upload
│       ├── services/
│       │   ├── auth
│       │   ├── project
│       │   ├── inquiry
│       │   └── upload
│       ├── validators/
│       │   ├── auth.schema.js
│       │   ├── project.schema.js
│       │   └── inquiry.schema.js
│       ├── middleware/
│       │   ├── auth.js
│       │   ├── validate.js
│       │   ├── rateLimit.js
│       │   ├── errorHandler.js
│       │   ├── notFound.js
│       │   ├── upload.js
│       │   ├── originCheck.js
│       │   ├── ApiError.js
│       │   ├── asyncHandler.js
│       │   ├── slugify.js
│       │   └── response.js
│       ├── utils/
│       ├── scripts/seed.js
│       ├── tests/
│       │   ├── auth
│       │   ├── projects
│       │   ├── inquiries
│       │   └── health
│       ├── .env.example
│       ├── package.json
│       ├── vitest.config.js
│       ├── .github/workflows/ci.yml
│       ├── package.json
│       ├── README.md
│       ├── .gitignore
│       ├── .editorconfig
│       └── .prettierrc
```

2.1 Dependencies

App	dependencies	devDependencies
client	react, react-dom, react-router-dom, framer-motion, @tanstack/react-query, axios, react-hook-form, @hookform/resolvers, zod, react-helmet-async, lucide-react, sonner, react-markdown, lenis, clsx, tailwind-merge	vite, @vitejs/plugin-react, tailwindcss, postcss, autoprefixer, eslint (+ react, react-hooks, jsx-a11y), prettier, vitest, jsdom, @testing-library/react, @testing-library/user-event, @testing-library/jest-dom
server	express, mongoose, zod, jsonwebtoken, bcryptjs, cookie-parser, cors, helmet, express-rate-limit, express-mongo-sanitize, compression, morgan, multer, cloudinary, nodemailer, dotenv	vitest, supertest, mongodb-memory-server, eslint, prettier

2.2 Scripts

```
root : "dev" concurrently "npm run dev --prefix server" "npm run dev --prefix client"
      "build" npm run build --prefix client
      "lint" / "test" run in both apps
client : dev (vite) | build (node scripts/generate-sitemap.js && vite build) | preview
        lint | test (vitest run)
server : dev (node --watch src/server.js) | start (node src/server.js)
        seed (node src/scripts/seed.js) | lint | test (vitest run)
```

3. Design System

3.1 Colour tokens

Token	Value	Usage
--bg-base	#07070B	Page background
--bg-elevated	#0D0D14	Navbar, footer, alternating sections
--surface	#12121B	Cards, inputs
--surface-hover	#181824	Card hover, table row hover
--border-subtle / --border	rgba(255, 255, 255, .06) / rgba(255, 255, 255, .10)	Dividers / card and input borders
--text	#F5F5F7	Headings and primary text
--text-2	#A1A1B5	Body copy, secondary text
--text-3	#6B6B80	Captions, placeholders (large text only)
--accent	#7C5CFF (hover #9277FF)	Primary actions, links, focus ring
--accent-2	#22D3EE	Eyebrow labels, highlights, gradient end
--accent-3	#F472B6	Aurora orb and rare gradient accents only
--success / --warning / --danger	#34D399 / #FBBF24 / #F87171	Form and status feedback
--gradient-brand	linear-gradient(135deg, #7C5CFF 0%, #22D3EE 100%)	Primary button, gradient text, gradient borders

3.2 Typography

Fonts from Google Fonts (preconnect, `display=swap`, subset latin): **Space Grotesk** 500/600/700 for display and headings, **Inter** 400/500/600 for body and UI, **JetBrains Mono** 400/500 for eyebrows and small labels. Provide system fallbacks for each.

Style	Size / line-height	Weight / tracking	Notes
display	clamp(2.75rem, 6vw, 5rem) / 1.05	700 / -0.03em	Hero headline, Space Grotesk
h2	clamp(2rem, 4vw, 3rem) / 1.1	600 / -0.02em	Section titles
h3	1.5rem / 1.3	600 / -0.01em	Card titles
body-lg	1.125rem / 1.7	400	Hero and section subtitles, colour <code>--text-2</code>
body	1rem / 1.7	400	Default
small	0.875rem / 1.5	400 / 500	Meta, table cells
eyebrow	0.75rem / 1	500 / 0.14em uppercase	JetBrains Mono, colour <code>--accent-2</code> , always above a section title

3.3 Layout, spacing and shape

Rule	Value
Container	max-width 1200px; horizontal padding 24px (20px under 640px); centred
Grid	12 columns, 24px gap; cards use 1 / 2 / 3 columns at sm / md / lg
Section padding	96px top and bottom from 1024px; 64px below
Spacing scale	4, 8, 12, 16, 24, 32, 48, 64, 96, 128 px (Tailwind default steps only)
Radius	sm 8px (inputs, chips) md 12px (buttons) lg 16px (cards) xl 24px (hero panels, featured) full (pills)
Breakpoints	sm 640 md 768 lg 1024 xl 1280
Card shadow	inset 0 1px 0 rgba(255,255,255,.04), 0 20px 40px -20px rgba(0,0,0,.6)
Focus ring	2px solid <code>--accent</code> with 2px offset on every interactive element (<code>:focus-visible</code>)
Icons	lucide-react, 20px inline / 24px feature, stroke width 1.75

3.4 Motion

Token	Value
Durations	micro 150ms (hover, press) base 300ms reveal 600ms
Easing	cubic-bezier(0.22, 1, 0.36, 1) for everything
Scroll reveal	opacity 0 to 1, y 24 to 0, once, viewport margin -80px, children stagger 80ms
Card hover	translateY(-4px), border to accent at 40%, spotlight glow fades in
Page transition	fade + 8px rise, 250ms, via AnimatePresence in PageTransition
Reduced motion	<code>prefers-reduced-motion: reduce</code> disables Lenis, orb floating, marquee, counters and reveal offsets (opacity only)

3.5 Effect recipes

- **Aurora:** container `absolute inset-0 overflow-hidden pointer-events-none`; three circles 500-700px, `blur-3xl`, violet / cyan / pink at 25-35% opacity, keyframe `float` 18-24s alternate using transform only. Max 3 blurred layers on any screen.
- **Grid mask:** background of two 1px lines (rgba white .04) at 64px, masked with `radial-gradient(ellipse at center, black 30%, transparent 75%)`.
- **Noise:** fixed full-screen div, inline SVG `feTurbulence` as data-URI background, opacity .035, `mix-blend-mode: overlay`, `pointer-events: none`, z-index above content but below modals.
- **Spotlight card:** `useSpotlight` sets `--mx` and `--my` on pointermove; card `::before` paints `radial-gradient(500px circle at var(--mx) var(--my), rgba(124,92,255,.18), transparent 40%)` and fades in on hover. Skip on touch devices.
- **Gradient border:** wrapper with 1px padding and gradient background, inner element with surface colour and radius minus 1px.
- **Primary button:** gradient background, white text, glow `0 8px 30px -8px rgba(124,92,255,.6)`, animated sheen sweep on hover, 2px press translate.
- **Marquee:** two identical tracks, `translateX(-50%)` linear infinite 30s, pause on hover, edge fade mask.
- **Counters:** `useCountUp` runs once when 40% visible, 1.4s ease-out, tabular numerals.
- **Scrollbar and selection:** thin dark scrollbar; `::selection` violet at 35% opacity. No custom cursor (accessibility).

3.6 Tailwind configuration (excerpt)

```
// client/tailwind.config.js
export default {
  content: ['./index.html', './src/**/*.js,jsx'],
  theme: {
    container: { center: true, padding: { DEFAULT: '1.25rem', sm: '1.5rem' },
      screens: { xl: '1200px' } },
    extend: {
      colors: {
        base: 'var(--bg-base)', elevated: 'var(--bg-elevated)', surface: 'var(--surface)',
        'surface-hover': 'var(--surface-hover)', border: 'var(--border)',
        'border-subtle': 'var(--border-subtle)', fg: 'var(--text)', 'fg-2': 'var(--text-2)',
        'fg-3': 'var(--text-3)', accent: 'var(--accent)', 'accent-2': 'var(--accent-2)',
        'accent-3': 'var(--accent-3)',
      },
      fontFamily: {
        display: ['"Space Grotesk"', 'ui-sans-serif', 'system-ui', 'sans-serif'],
        sans: ['Inter', 'ui-sans-serif', 'system-ui', 'sans-serif'],
        mono: ['"JetBrains Mono"', 'ui-monospace', 'monospace'],
      },
      borderRadius: { sm: '8px', md: '12px', lg: '16px', xl: '24px' },
      transitionTimingFunction: { out: 'cubic-bezier(0.22, 1, 0.36, 1)' },
      keyframes: {
        float: { '0%': { transform: 'translate3d(0,0,0)' },
          '100%': { transform: 'translate3d(40px,-30px,0)' } },
        marquee: { to: { transform: 'translateX(-50%)' } },
        sheen: { to: { transform: 'translateX(220%)' } },
      },
      animation: { float: 'float 20s ease-in-out infinite alternate',
        marquee: 'marquee 30s linear infinite' },
    },
  },
};
```

3.7 Global CSS tokens (excerpt)

```
/* client/src/styles/index.css */
@tailwind base; @tailwind components; @tailwind utilities;
:root {
  --bg-base:#07070B; --bg-elevated:#0D0D14; --surface:#12121B; --surface-hover:#181824;
  --border:rgba(255,255,255,.10); --border-subtle:rgba(255,255,255,.06);
  --text:#F5F5F7; --text-2:#A1A1B5; --text-3:#6B6B80;
  --accent:#7C5CFF; --accent-2:#22D3EE; --accent-3:#F472B6;
  --gradient-brand:linear-gradient(135deg,#7C5CFF 0%,#22D3EE 100%);
  color-scheme: dark;
}
html { background: var(--bg-base); scroll-behavior: auto; }
body { @apply bg-base text-fg-2 font-sans antialiased; }
:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
@layer components {
  .eyebrow { @apply font-mono text-xs uppercase tracking-[.14em] text-accent-2; }
  .text-gradient { background: var(--gradient-brand); -webkit-background-clip: text;
    background-clip: text; color: transparent; }
}
@media (prefers-reduced-motion: reduce) { *,*::before,*::after { animation: none !important;
  transition-duration: .01ms !important; } }
```

3.8 Consistency rules (enforced)

- Never hard-code colours, font families or shadows in components; use tokens and Tailwind theme names.
- Every page starts with the same shell: **Navbar** > **main** > sections > **Footer**; the first section on inner pages is a compact page hero (eyebrow, h1, subtitle).
- Section rhythm alternates **bg-base** and **bg-elevated** only where it aids separation; dividers use **border-subtle**.
- Titles use the pattern eyebrow + h2 (with one gradient word max) + body-lg subtitle, max width 640px, centred or left per section spec.
- Buttons: primary (gradient) for the single main action per view, secondary (outlined) for alternatives, ghost for tertiary. Never two primary buttons in one viewport.
- Interactive cards have hover lift, spotlight and focus-visible states; non-interactive cards do not lift.
- All images have explicit width, height (or aspect-ratio), alt text and lazy loading (except the hero).
- Copy tone: confident, short sentences, no jargon. Buttons are verbs ("Start a project", "View our work").

4. Component Library

4.1 UI primitives (components/ui)

Component	Props	Behaviour
Button	variant: primary secondary ghost; size: sm md lg; to / href / onClick; loading; iconRight; disabled	Renders Link , a or button ; spinner when loading; sheen on hover; min touch target 44px
Card	interactive; spotlight; gradientBorder; className; children	Surface, border, radius lg; spotlight and lift only when interactive
Badge	tone: neutral accent success warning; children	Small pill, mono 12px
Container / Section	Section: id, tone: base elevated, className	Applies max width, padding and rhythm; provides anchor id with scroll-margin-top 96px
SectionHeader	eyebrow, title (node), subtitle, align: left center	Wraps in Reveal ; renders h2
Input / Textarea / Select	label, name, error, hint, register props	Label above, error text below with aria-describedby , danger border on error
Accordion	items: [{q, a}]; single-open	Button headers with aria-expanded , animated height, keyboard operable
FilterChips	options, value, onChange	Radio-group semantics, arrow-key navigation, animated active pill (layoutId)
Skeleton	className	Shimmering placeholder using surface tokens
GradientText	children	Inline span with .text-gradient
Reveal	delay, y, once, as	Framer Motion whileInView wrapper; opacity-only when reduced motion
Marquee	items, speed, pauseOnHover	Duplicated track with mask edges
Counter	to, suffix, prefix, duration	Uses useCountUp ; announces final value to screen readers
Modal	open, onClose, title, children	Focus trap, escape to close, scroll lock (admin confirmations and inquiry view)

4.2 Layout components (components/layout)

Component	Specification
Navbar	Fixed top. Height 72px, shrinks to 64px after 24px scroll, background rgba(7,7,11,.7) with backdrop-blur-xl and bottom border-subtle. Left: wordmark (brand name + gradient dot). Centre: Home, Work, About, Pricing, Contact with animated underline (layoutId) on the active route. Right: primary button "Start a project". Below md: hamburger opens a full-screen overlay menu with staggered links; body scroll locked; closes on route change and Escape.
Footer	Four columns: brand + tagline + social icons; Navigate; Services; Contact (email, phone, city). Bottom bar with copyright and "Back to top". Oversized faint wordmark (text-fg/5, clipped at bottom) as a decorative element. Top border-subtle.
PageTransition	AnimatePresence wrapper keyed by pathname; fade + 8px rise; scrolls to top on route change (ScrollToTop), respects hash anchors.
Aurora	Absolute background layer with orbs and grid mask (see 3.5). Props: intensity low medium high.
NoiseOverlay	Single fixed grain layer mounted once in App.

4.3 Feature components

Component	Specification
ProjectCard	Interactive Card linking to <code>/projects/:slug</code> . Cover 16:10 with lazy image (hover scale 1.04, 500ms), category Badge top-left, title, one-line summary (2-line clamp), up to 3 tech chips, arrow icon that slides in on hover. Featured projects span 2 columns on lg with gradient border.
ProjectGrid	Handles loading (6 skeleton cards), error (message + Retry button), empty ("New work coming soon") and populated states. Grid 1 / 2 / 3 columns.
PricingCard	name, price, period, description, features[], popular, cta. Popular card has gradient border, "Most popular" Badge, slightly larger padding and primary button; others use secondary.
StatCard	value, suffix, label; Counter + label; separated by vertical dividers on desktop.
ServiceTile	icon, title, description, size (sm wide tall) for the bento grid; interactive Card.
TeamCard	photo placeholder (initials on gradient), name, role, social icons.
TestimonialCard	quote, author, role, company, avatar initials; quote mark icon.
ProcessStep	index (mono "01"), title, description; vertical line connector with animated fill on scroll.
ContactForm	Fields and rules in section 5.5; React Hook Form + Zod; submit button loading state; success panel replaces form; sonner toast on error.

5. Page Specifications

Every public page sets its own SEO tags via `<Seo />` (title, description, canonical path, Open Graph image). The default title template is `%s | {brand.name}`.

5.1 Home (/)

Navbar (fixed)	Global
Hero + aurora + grid	Headline, subtitle, 2 CTAs, trust chips, floating project preview cards
Tech marquee	Logos / names of technologies, infinite scroll
Stats bar	4 animated counters
Services bento grid	Six tiles, mixed sizes
OUR WORK (id="work")	Published projects from API, filter chips, load more
Process timeline	Five steps
Testimonials	Three cards
Pricing teaser	Three compact plans + link to /pricing
CTA banner	Gradient panel with aurora, "Start a project"
Footer	Global

Section	Content and behaviour
Hero	Eyebrow "WEB DESIGN & DEVELOPMENT STUDIO". H1 (display): "We build websites that make brands unforgettable ." (last word gradient). Subtitle: "{Brand} is a full-service web studio crafting fast, beautiful, conversion-focused websites and web apps for ambitious businesses." Buttons: primary "Start a project" (/contact), secondary "View our work" (#work). Trust chips from config: "50+ projects delivered", "4.9/5 client rating", "3-week average delivery". Right side (lg+): 2-3 tilted glass cards previewing featured projects with slow parallax float. Aurora intensity high.
Tech marquee	React, Node.js, MongoDB, Next.js, TypeScript, Tailwind, Figma, Stripe, AWS, Vercel, PostgreSQL, Shopify (text with lucide or simple SVG marks, monochrome at 60% opacity, full colour on hover).
Stats	50+ Projects delivered 30+ Happy clients 4.9 Average rating 3 wks Average delivery. Values from content/stats.js .
Services	Eyebrow "WHAT WE DO", title "Everything your brand needs online ". Bento with six ServiceTiles (see 6.1). Each tile is interactive with spotlight.

Section	Content and behaviour
Our Work	Eyebrow "SELECTED WORK", title "Projects we are proud of ", subtitle "Real products shipped for real businesses." FilterChips: All, Websites, Web Apps, E-commerce, Branding (chips generated from categories present in data). Data via <code>useProjects({ category, limit: 6 })</code> ; "Load more" button appends the next page. Featured project spans two columns. Ends with a secondary button "Discuss your project".
Process	Eyebrow "HOW WE WORK", title "A clear path from idea to launch ". Five ProcessSteps (see 6.1).
Testimonials	Three TestimonialCards from content; on md they sit in a 3-column grid, on mobile a horizontal snap scroller.
Pricing teaser	Compact three-plan row (name, from-price, one-line summary) + "See full pricing" ghost button to /pricing.
CTA banner	Full-width xl-radius panel with gradient border and aurora. Title "Have a project in mind?" Subtitle "Tell us about it - we reply within one business day." Primary button "Start a project".

5.2 About (/about)

Section	Content and behaviour
Page hero	Eyebrow "ABOUT US", h1 "A small team obsessed with craft ", subtitle about the agency mission. Smaller aurora.
Story	Two columns: left three short paragraphs (founded, approach, promise); right a glass card with founding year, team size and location from config.
Values	Four ServiceTile-style cards: Quality first, Radical transparency, Speed with care, Long-term partnership.
Stats	Reuse the stats bar component.
Team	Four TeamCards (placeholder names and roles: Founder & Lead Engineer, Design Director, Full-Stack Developer, Project Manager).
Milestones	Vertical timeline with four entries (year, title, one line). Alternating sides on lg, single column on mobile.
Stack	Grid of technology chips grouped Frontend / Backend / Design / DevOps from <code>content/techStack.js</code> .
CTA banner	Same component as Home.

5.3 Pricing (/pricing)

Plan	Starter	Business (Most popular)	Enterprise
Price (sample)	\$499 one-time	\$1,499 one-time	Custom
Best for	Personal brands, small local businesses	Growing companies needing a full site with content management	Web apps, integrations, complex platforms
Pages / scope	Up to 5 pages	Up to 10 pages	Custom scope
Design	Template-based custom styling	Fully custom design	Custom design system
Features	Responsive layout, contact form, basic SEO, social links	Everything in Starter + admin panel / CMS, animations, analytics, blog-ready structure	Everything in Business + custom backend, third-party integrations, auth, dashboards
Revisions	1 round	3 rounds	Agreed in contract
Delivery	~1 week	2 to 3 weeks	Milestone based
Support	14 days	60 days	SLA available
CTA	Get started	Get started	Talk to us

- **Page hero:** eyebrow "PRICING", h1 "Simple, transparent **pricing**", subtitle "Fixed-scope packages with no surprises. Need something bigger? Let us talk."
- **Plan cards:** three PricingCards; middle is popular. Data from `content/pricing.js`; price strings and currency are configuration only.
- **Comparison table:** rows = features, columns = plans, check / dash icons with accessible labels; sticky first column and horizontal scroll on mobile.
- **Add-ons (Card grid):** Extra page, Logo and brand kit, Content writing, Monthly maintenance, Hosting and domain setup, SEO audit (each with a "from" price).
- **FAQ (Accordion, 6 items):** How long does a project take? What do I need to provide? Do you offer maintenance? Who owns the code and content? Can you redesign an existing site? What are the payment terms (50% upfront, 50% on launch)?
- **CTA banner:** same component.

5.4 Project detail (/projects/:slug)

- **Header:** back link "All work", category Badge, h1 title, summary, meta grid (Client, Year, Category, Industry) and buttons "Visit live site" (external, `rel="noopener noreferrer"`) if `liveUrl` exists.
- **Hero image:** full-width cover in a bordered xl-radius frame with soft glow.
- **Body:** description rendered with react-markdown (headings styled to tokens, links accent, no raw HTML). Two-column layout on lg: content left, sticky sidebar right (tech stack chips, results).
- **Results:** up to four StatCards from `results[]` (label + value such as "+120%" "organic traffic").
- **Gallery:** responsive grid of gallery images; click opens a lightbox Modal with arrow-key navigation.
- **Testimonial:** single quote block if present.
- **Footer:** "Next project" card (next published project by order) and CTA banner.
- **States:** skeleton while loading; branded not-found panel with link home if API returns 404; SEO with project title, summary, cover as OG image and CreativeWork JSON-LD.

5.5 Contact (/contact)

Two columns on lg: left the ContactForm inside a Card, right contact details (email, phone, city, response-time promise "We reply within one business day") and a small "What happens next" three-step list.

Field	Type	Rules
name	text	Required, 2-80 chars
email	email	Required, valid email, max 120
company	text	Optional, max 120
projectType	select	Required: New website Redesign Web app E-commerce Other
budget	select	Required: Under \$1k \$1k-\$3k \$3k-\$10k \$10k+ Not sure
message	textarea	Required, 10-2000 chars, live character count
website (honeypot)	hidden text	Must stay empty; visually hidden with <code>tabindex=-1</code> and <code>aria-hidden</code>

On success: the form is replaced by a success panel ("Thanks - we will be in touch within one business day") with a "Send another" link. On failure: sonner error toast and the form keeps its values. Below the button: "By sending this form you agree to be contacted about your project."

5.6 404 and global states

- **404:** large gradient "404", short line, buttons "Go home" and "Contact us".
- **Error boundary:** friendly full-page fallback with reload button; logs to console in dev.
- **Offline / API down:** ProjectGrid shows the error state with Retry; other pages remain functional because their content is static.

6. Static Content Definitions

6.1 Content files (client/src/content)

File	Shape and default content
services.js	{ { icon, title, description, size } } Website Design & Development (wide); Web Apps & Dashboards; E-commerce Stores; UI/UX Design (tall); SEO & Performance; Maintenance & Support
process.js	{ { step, title, description } } 01 Discover (goals, audience, scope); 02 Design (wireframes, UI, prototype); 03 Develop (build, integrate, test); 04 Launch (deploy, monitor, train); 05 Support (updates, growth)
stats.js	{ { value, suffix, label } } 50+ Projects delivered; 30+ Happy clients; 4.9 Average rating; 3 (weeks) Average delivery
testimonials.js	3 items: { quote, author, role, company }. Write plausible placeholder quotes and mark clearly as sample in a code comment
team.js	4 items: { name, role, initials, links } (placeholders)
values.js	4 items: { icon, title, description }
techStack.js	{ frontend[], backend[], design[], devops[] } string arrays
pricing.js	{ currency, plans[], comparison[], addOns[] } matching section 5.3
faq.js	{ { q, a } } six entries from 5.3

6.2 Site config (client/src/config/site.js)

```
export const brand = { name: 'Studio Name', tagline: 'Websites that make brands unforgettable',
  url: 'https://example.com', email: 'hello@example.com', phone: '+1 555 000 0000',
  city: 'City, Country', founded: 2021, teamSize: 8 };
export const nav = [ { label: 'Home', to: '/' }, { label: 'Work', to: '/#work' },
  { label: 'About', to: '/about' }, { label: 'Pricing', to: '/pricing' },
  { label: 'Contact', to: '/contact' } ];
export const socials = [ { name: 'GitHub', href: '#' }, { name: 'LinkedIn', href: '#' },
  { name: 'Instagram', href: '#' }, { name: 'X', href: '#' } ];
export const heroChips = ['50+ projects delivered', '4.9/5 client rating', '3-week average delivery'];
```

7. Data Models (Mongoose)

7.1 Project

```
const projectSchema = new Schema({
  title: { type: String, required: true, trim: true, maxlength: 120 },
  slug: { type: String, required: true, unique: true, lowercase: true, index: true },
  summary: { type: String, required: true, maxlength: 240 },
  description: { type: String, default: '' }, // markdown case study
  category: { type: String, enum: ['website', 'web-app', 'ecommerce', 'branding'], required: true },
  industry: { type: String, default: '' },
  clientName: { type: String, default: '' },
  year: { type: Number, min: 2000, max: 2100 },
  techStack: { type: [String], default: [] },
  coverImage: { url: String, publicId: String, alt: String },
  gallery: [{ url: String, publicId: String, alt: String }],
  liveUrl: { type: String, default: '' },
  results: [{ label: String, value: String }], // max 4
  testimonial: { quote: String, author: String, role: String },
  featured: { type: Boolean, default: false },
  status: { type: String, enum: ['draft', 'published'], default: 'draft', index: true },
  publishedAt: Date, // set when first published
  order: { type: Number, default: 0 }, // lower = earlier
}, { timestamps: true });
projectSchema.index({ status: 1, order: 1, publishedAt: -1 });
```

7.2 Inquiry and User

Model	Fields	Indexes / notes
Inquiry	name, email, company, projectType (enum), budget (enum), message, status (new read replied archived, default new), ip (string), userAgent (string), createdAt	Index on status + createdAt desc. Never returned by public endpoints.
User	email (unique, lowercase), passwordHash, name, role (admin), lastLoginAt	passwordHash has <code>select: false</code> . <code>toJSON</code> strips it.

Slugs are generated from the title with `slugify` (lowercase, hyphens, ASCII) and made unique by appending `-2`, `-3` on collision. Slugs are editable in the admin form but stay unique.

8. API Specification

8.1 Conventions

- Base path `/api`. JSON only (multipart for uploads). Responses use one envelope:

```
success: { "success": true, "data": <any>, "meta": { "page":1, "limit":12, "total":37, "totalPages":4 } }  
error:   { "success": false, "error": { "code": "VALIDATION_ERROR", "message": "Invalid input",  
                                         "details": [ { "path": "email", "message": "Invalid email" } ] } }
```

HTTP	Error code	When
400	<code>VALIDATION_ERROR</code> / <code>BAD_REQUEST</code>	Zod failure, malformed input, bad upload
401	<code>UNAUTHORIZED</code>	Missing, invalid or expired cookie; wrong login
403	<code>FORBIDDEN</code>	Origin check failed on a state-changing admin request
404	<code>NOT_FOUND</code>	Unknown route, unknown or unpublished slug
409	<code>CONFLICT</code>	Duplicate slug or unique key
429	<code>RATE_LIMITED</code>	Rate limit exceeded (include <code>Retry-After</code>)
500	<code>SERVER_ERROR</code>	Unexpected error; generic message, no stack trace in production

8.2 Endpoints

Method + path	Auth	Purpose and notes
GET /api/health	Public	Returns { status: "ok", uptime, db: "connected" }; used by uptime monitor and Render
GET /api/projects	Public	Published projects only. Query: <code>category</code> , <code>featured</code> (true), <code>page</code> (1), <code>limit</code> (12, max 50). Sorted by <code>featured</code> desc, <code>order</code> asc, <code>publishedAt</code> desc. Response cards omit <code>description</code> . Header <code>Cache-Control: public, max-age=60, stale-while-revalidate=300</code>
GET /api/projects/:slug	Public	One published project with full fields; 404 if draft or missing
POST /api/inquiries	Public	Body per 5.5. Rate limit 5 per hour per IP. Honeypot filled: return 201 without saving. Saves, then sends email (non-blocking). Returns 201 { id }
POST /api/auth/login	Public	Body { email, password }. Rate limit 5 per 15 min per IP. Sets httpOnly cookie <code>token</code> . Returns { user }
POST /api/auth/logout	Admin	Clears cookie; 204
GET /api/auth/me	Admin	Returns current user or 401
GET /api/admin/stats	Admin	Counts: projects total, published, drafts, inquiries new; latest 5 inquiries
GET /api/admin/projects	Admin	All projects incl. drafts. Query: <code>status</code> , <code>q</code> (title search), <code>page</code> , <code>limit</code>
GET /api/admin/projects/:id	Admin	Single project by id (for edit form)
POST /api/admin/projects	Admin	Create (defaults to draft). Validates with project schema
PUT /api/admin/projects/:id	Admin	Replace editable fields; slug uniqueness enforced
PATCH /api/admin/projects/:id/status	Admin	Body { status }. Sets <code>publishedAt</code> on first publish
DELETE /api/admin/projects/:id	Admin	Deletes project and best-effort deletes its Cloudinary images
POST /api/admin/uploads	Admin	multipart field <code>image</code> ; jpeg, png, webp, avif; max 5 MB; returns { url, publicId, width, height }
DELETE /api/admin/uploads/:publicId	Admin	Remove an uploaded image (publicId URL-encoded)
GET /api/admin/inquiries	Admin	Paginated, filter by <code>status</code>
PATCH /api/admin/inquiries/:id	Admin	Body { status }
DELETE /api/admin/inquiries/:id	Admin	Hard delete

8.3 Validation rules (Zod)

Schema	Rules
<code>inquiry</code>	name 2-80; email valid, max 120; company max 120 optional; <code>projectType</code> and <code>budget</code> must match the enums in 5.5; message 10-2000; website (honeypot) max 0 chars. Trim all strings.
<code>login</code>	email valid; password 8-128 chars
<code>project</code>	title 3-120; summary 10-240; category enum; year optional int; <code>techStack</code> max 12 items each max 30 chars; results max 4 items; gallery max 12; <code>liveUrl</code> empty or valid http(s) URL; status enum; order integer; <code>coverImage.url</code> required to publish (enforced in the status endpoint)
<code>query</code>	page and limit coerced to integers with defaults and caps; category restricted to the enum

9. Server Implementation Details

9.1 app.js middleware order

```

app.set('trust proxy', 1); // behind Render / Vercel proxy
app.use(helmet());
app.use(cors({ origin: env.CLIENT_ORIGINS, credentials: true })); // allowlist array
app.use(compression());
app.use(express.json({ limit: '100kb' }));
app.use(cookieParser());
app.use(mongoSanitize());
if (env.NODE_ENV !== 'test') app.use(morgan(env.NODE_ENV === 'production' ? 'combined' : 'dev'));
app.use('/api', globalLimiter); // 100 requests / 15 min / IP
app.use('/api', originCheck); // POST/PUT/PATCH/DELETE: Origin must be allowed
app.use('/api', routes);
app.use(notFound);
app.use(errorHandler);

```

9.2 Authentication

- Login: find user (`select +passwordHash`), compare with `bcrypt`; if the user does not exist compare against a dummy hash to keep timing constant; always answer "Invalid email or password".
- Token: `jwt.sign({ sub: user.id, role }, JWT_SECRET, { expiresIn: JWT_EXPIRES_IN })` (default 1d).
- Cookie: name `token`, `httpOnly: true`, `secure: NODE_ENV === "production"`, `sameSite: "lax"`, `path: "/"`, `maxAge` 1 day.
- `requireAuth` middleware reads the cookie, verifies the token, loads the user and sets `req.user`; failures throw `ApiError(401)`.
- CSRF posture: `SameSite=Lax` cookie, JSON-only bodies, CORS allowlist and the `originCheck` middleware on state-changing requests.

9.3 Uploads, email, errors, logging

- **Upload:** `multer` memory storage (5 MB, single file); reject unsupported MIME types; stream to Cloudinary folder `agency/projects` with `resource_type: "image"`; return secure URL and `publicId`. Client builds delivery URLs with `f_auto,q_auto,w_{n}` transforms via a `cdn(url, width)` helper in `lib/utils.js`.
- **Email:** `mail.service.js` wraps `Nodemailer`. `sendInquiryNotification(inquiry)` sends text + simple HTML to `MAIL_TO` with all fields and a reply-to of the visitor. Wrap in try/catch and log; never throw into the request. Skip silently when SMTP env vars are absent (development).
- **Errors:** `ApiError(status, code, message, details)`. Central `errorHandler` maps `ZodError` to 400, `Mongo duplicate key (11000)` to 409, `JsonWebTokenError` and `TokenExpiredError` to 401, `MulterError` to 400, everything else to 500. Log the stack server-side only.

- **Startup:** validate env with Zod, connect to Mongo (fail fast on error), then listen. Handle SIGTERM by closing the server and the Mongo connection. Health route reports `mongoose.connection.readyState`.
- **Seed** (`npm run seed`): idempotent. Upserts the admin from `ADMIN_EMAIL` and `ADMIN_PASSWORD`; inserts six sample published projects only when the collection is empty (see 9.4). Refuses to wipe data in production.

9.4 Seed projects (placeholders)

Title	Category	Featured	Tech
Lumen Coffee Roasters	ecommerce	Yes	React, Node.js, Stripe, MongoDB
Atlas Fintech Dashboard	web-app	Yes	React, Express, PostgreSQL, Chart.js
Northwind Legal	website	No	React, Tailwind, Framer Motion
FitPulse Coaching App	web-app	No	React, Node.js, MongoDB, JWT
Verdant Interiors	website	No	React, Node.js, Cloudinary
Orbit SaaS Landing	branding	No	React, Tailwind, Lottie

Each seed project has a 3 to 4 paragraph markdown description (challenge, solution, outcome), 2 results (for example "+120%" "organic traffic"), `year` 2024-2026 and a generated placeholder cover. Create six distinct gradient SVG mockups in `client/public/placeholders/project-1.svg` to `project-6.svg` (browser-frame style with abstract UI blocks in the brand palette) and reference them as `/placeholders/project-N.svg`.

9.5 Environment variables

Variable	App	Example / notes
<code>NODE_ENV</code>	server	development test production
<code>PORT</code>	server	5000 (Render supplies its own)
<code>MONGODB_URI</code>	server	<code>mongodb+srv://user:pass@cluster/agency?retryWrites=true&w=majority</code>
<code>JWT_SECRET</code>	server	Random string, 48+ chars
<code>JWT_EXPIRES_IN</code>	server	1d
<code>CLIENT_ORIGINS</code>	server	Comma list: <code>http://localhost:5173,https://your-domain.com</code>
<code>CLOUDINARY_CLOUD_NAME</code> / <code>_API_KEY</code> / <code>_API_SECRET</code>	server	From Cloudinary dashboard
<code>SMTP_HOST</code> / <code>SMTP_PORT</code> / <code>SMTP_USER</code> / <code>SMTP_PASS</code>	server	Any SMTP provider
<code>MAIL_FROM</code> / <code>MAIL_TO</code>	server	"Studio <no-reply@domain>" / inbox that receives inquiries
<code>ADMIN_EMAIL</code> / <code>ADMIN_PASSWORD</code>	server	Used only by the seed script; password 12+ chars
<code>VITE_API_BASE_URL</code>	client	<code>/api</code> (proxied in dev and on Vercel)
<code>VITE_SITE_URL</code>	client	<code>https://your-domain.com</code> (canonical URLs, sitemap)

10. Client Implementation Details

10.1 Routing

Path	Component	Loading
/	pages/Home	Eager (critical path)
/about, /pricing, /contact, /projects/:slug	pages/*	<code>React.lazy</code> + Suspense with skeleton
/admin/login	pages/admin/Login	Lazy, separate chunk
/admin/*	AdminLayout inside <code>ProtectedRoute</code>	Lazy admin chunk; redirects to /admin/login when <code>GET /auth/me</code> returns 401
/styleguide	pages/StyleGuide	Dev only (<code>import.meta.env.DEV</code>)
*	pages/NotFound	Eager

10.2 Data layer

- `lib/api.js`: Axios instance, `baseUrl = import.meta.env.VITE_API_BASE_URL || "/api"`, `withCredentials: true`, response interceptor normalises errors to `{ message, code, details, status }`.
- `lib/queryClient.js`: `staleTime` 5 minutes, `retry` 1, `refetchOnWindowFocus` false for public queries.
- Hooks: `useProjects(params)` (infinite query for Load more), `useProject(slug)`, `useSubmitInquiry()` (mutation), admin hooks per resource with cache invalidation after mutations.
- `AuthContext` wraps `GET /auth/me`, `login`, `logout`; exposes `user`, `isLoading`. Only the admin chunk consumes it.

10.3 SEO

- `<Seo title description path image type jsonLd />` built on react-helmet-async; emits title, description, canonical, og:title, og:description, og:image, og:url, twitter:card. Admin routes emit `noindex, nofollow`.
- Global JSON-LD `Organization` and `WebSite` on Home; `CreativeWork` (name, description, image, creator, dateCreated) on project detail.
- `public/robots.txt`: allow all, `Disallow: /admin`, sitemap link. `scripts/generate-sitemap.js` writes `public/sitemap.xml` with static routes plus published project slugs fetched from the absolute API URL in env `SITEMAP_API_URL`; on failure it writes static routes only and exits 0.
- `index.html`: theme-color #07070B, favicon.svg, font preconnect, an OG default image 1200x630 (generate a branded one).

10.4 Performance and accessibility checklist

- Route-level code splitting; admin bundle never loaded for public visitors; dynamic import Framer Motion features with `LazyMotion` and `domAnimation`.
- Fonts: preload the two critical woff2 files, `font-display: swap`; limit to the weights listed in 3.2.
- Images: `loading="lazy"` and `decoding="async"` except the hero; explicit `width` and `height`; Cloudinary `srcset` at 480, 800, 1200.
- Animate only `transform` and `opacity`; cap simultaneous blurred layers at three; pause marquee and orbs when the tab is hidden.
- Semantic landmarks (`header`, `nav`, `main`, `footer`), one h1 per page, skip-to-content link, `aria-current="page"` on the active link, labelled form controls, `aria-live` region for form status.

- Colour contrast verified: body text --text-2 on --bg-base is above 7:1; --text-3 is used only for large or decorative text.
- Keyboard: menu, chips, accordion, modal and lightbox fully operable; visible focus everywhere.

11. Admin Panel Specification

The admin UI reuses design tokens and UI primitives but is deliberately calm: no aurora, no noise, no page transitions, dense tables. Layout: left sidebar (Dashboard, Projects, Inquiries, View site, Logout), top bar with page title, content area max width 1200px.

Page	Specification
Login	Centred Card, email and password, show/hide password, loading state, generic error message. Redirects to /admin on success.
Dashboard	Four stat cards (Total projects, Published, Drafts, New inquiries) and a "Latest inquiries" list (name, project type, date, status Badge) linking to the inbox.
Projects	Table: cover thumbnail, title, category, status Badge, featured star, updated date, actions (Edit, Publish/Unpublish, Delete). Search by title, filter by status, pagination. Delete confirms in a Modal. "New project" primary button.
Project form	Sections: Basics (title, slug auto-generated and editable, summary, category, industry, client, year, live URL); Media (cover uploader with preview and alt text, gallery uploader up to 12 with remove); Content (markdown textarea with live preview tab); Details (tech stack chips input, results repeater up to 4, testimonial fields, featured switch, order number); footer bar with "Save draft" and "Publish". Warns on unsaved changes. Publish is disabled until title, summary, category and cover exist.
Inquiries	Table: date, name, email, type, budget, status. Row click opens a Modal with the full message, mailto reply button, and a status select (new, read, replied, archived). Delete with confirmation.

12. Testing and QA

12.1 Automated tests (Vitest)

Scope	Cases
Server: health	Returns ok and db state
Server: auth	Login success sets cookie; wrong password 401; protected route without cookie 401; logout clears cookie
Server: projects	Public list returns only published; category filter; pagination meta; draft slug 404; admin create, update, publish sets publishedAt, delete; duplicate slug handled
Server: inquiries	Valid body 201 and saved; invalid body 400 with details; honeypot returns 201 without saving; rate limit returns 429 on the sixth request
Client: UI	Button variants and loading state; Accordion keyboard toggle; FilterChips selection; PricingCard popular styling
Client: features	ContactForm validation messages, success panel, API error toast (Axios mocked); ProjectGrid loading, empty, error and populated states; route smoke test for all public routes

Server tests use [mongodb-memory-server](#) and Supertest and must not need external services. Target coverage: 80% on server services and controllers.

12.2 Manual QA checklist

- Viewports 320, 375, 768, 1024, 1440, 1920: no horizontal scroll, nothing clipped, tap targets at least 44px.
- Keyboard-only walkthrough of every page; screen-reader spot check of navbar, form and accordion.

- Reduced-motion enabled: no floating orbs, marquee, counters or smooth scroll.
- Lighthouse (mobile) on Home, Pricing and a project page: Performance ≥ 90 , Accessibility ≥ 95 , SEO ≥ 95 , Best Practices ≥ 95 .
- Browsers: Chrome, Edge, Firefox, Safari, iOS Safari.
- Contact form end to end: record in database and email received; API failure shows toast and keeps input.

13. Coding Standards Snapshot

- Controllers are thin (parse, call service, respond); services hold logic and DB access; models hold schema and indexes only.
- Wrap async handlers with `asyncHandler`; throw `ApiError`; never `res.send` from services.
- Use `.lean()` for read queries; select only needed fields for list endpoints; always paginate lists.
- No secrets, tokens or personal data in logs; log method, path, status and duration only.
- React: functional components, hooks only, no prop drilling beyond two levels (use composition or context), memoise only where profiling shows benefit.
- Keep components under about 200 lines; extract sections into `components/sections/`.

14. Build Phases and Acceptance Criteria

Phase	Tasks	Acceptance criteria
P0 Scaffold	Create monorepo, both apps, ESLint, Prettier, editorconfig, env validation, root scripts, CI workflow, README skeleton	<code>npm run dev</code> starts both apps; lint and tests run green in CI
P1 Design system	Tokens, Tailwind config, fonts, UI primitives, Navbar, Footer, Aurora, NoiseOverlay, Lenis, PageTransition, dev-only /styleguide showing every component and state	Style guide renders all components; navbar and footer responsive; no hard-coded colours; reduced-motion works
P2 Backend core	Config, db, models, middleware, health, public projects API, inquiries, mail service, seed, tests	All server tests pass; seed creates admin and 6 projects; curl of <code>/api/projects</code> returns envelope
P3 Public pages	Home (all sections), About, Pricing, Contact, ProjectDetail, NotFound, content files, SEO component, placeholder SVGs	All pages match section 5; Home lists seeded projects with working filters and Load more; contact form saves an inquiry
P4 Admin	Auth endpoints, AuthContext, ProtectedRoute, admin layout, dashboard, project CRUD with uploads, inquiry inbox, admin API tests	Admin can log in, create, publish and unpublish a project that appears and disappears on Home; images upload to Cloudinary
P5 Polish	Animations, empty and error states, SEO tags, sitemap, robots, JSON-LD, accessibility fixes, performance tuning	Meets NFR targets locally with Lighthouse; keyboard and reduced-motion checks pass
P6 Quality	Complete automated tests, manual QA checklist, bug fixes, README finalised	CI green; QA checklist ticked; no console errors or warnings
P7 Ship	Deploy per section 15, connect domain, production smoke test, handover note	Live URLs work; smoke tests pass; owner receives handover note

15. Deployment Runbook

Agent note: steps 1 to 4 require the owner to create accounts and copy secrets. Prepare `.env.example`, `vercel.json`, `render.yaml` (optional) and the CI workflow first, then present this list to the owner as a checklist. Continue with steps 5 onward as soon as the values are provided.

#	Step	Details
1	GitHub	Push the monorepo to a private repository on <code>main</code>
2	MongoDB Atlas	Create a free M0 cluster; create a database user; Network Access: allow 0.0.0.0/0 (Render free instances have dynamic IPs; rely on a strong password); copy the connection string into <code>MONGODB_URI</code> with database name <code>agency</code>
3	Cloudinary	Create a free account; copy cloud name, API key and secret
4	SMTP email	Use any provider (Brevo, Resend SMTP, Gmail app password); copy host, port, user, password; choose <code>MAIL_TO</code>
5	Render (API)	New Web Service from the repo; Root Directory <code>server</code> ; Build <code>npm ci</code> ; Start <code>npm start</code> ; Node 20; Health Check Path <code>/api/health</code> ; add all server env vars; set <code>NODE_ENV=production</code> ; <code>CLIENT_ORIGINS</code> = final frontend URL(s)
6	Seed	Once the service is live, run <code>npm run seed</code> (Render Shell or locally with the production <code>MONGODB_URI</code>); then log in and change nothing else
7	Vercel (client)	Import the repo; Root Directory <code>client</code> ; Framework Vite; Build <code>npm run build</code> ; Output <code>dist</code> ; env <code>VITE_API_BASE_URL=/api</code> , <code>VITE_SITE_URL</code> , <code>SITEMAP_API_URL</code> ; add <code>vercel.json</code> (below) with your Render URL
8	Domain and HTTPS	Add the custom domain in Vercel; set DNS records; wait for the certificate; update <code>CLIENT_ORIGINS</code> on Render and redeploy the API
9	Monitoring	Add an uptime monitor (for example UptimeRobot) pinging <code>https://your-domain/api/health</code> every 5-10 minutes; this also prevents free-tier sleep
10	Smoke test	See checklist below; record results in the handover note

```
// client/vercel.json (order matters: API rewrite first, SPA fallback second)
{
  "rewrites": [
    { "source": "/api/:path*", "destination": "https://YOUR-API.onrender.com/api/:path*" },
    { "source": "/(.*)", "destination": "/index.html" }
  ],
  "headers": [
    { "source": "/assets/(.*)",
      "headers": [{ "key": "Cache-Control", "value": "public, max-age=31536000, immutable" }] }
  ]
}
```

```
# .github/workflows/ci.yml
name: CI
on:
  push: { branches: [main] }
  pull_request:
jobs:
  build-test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20, cache: npm, cache-dependency-path: '**/package-lock.json' }
      - run: npm ci --prefix server
      - run: npm ci --prefix client
      - run: npm run lint --prefix server
      - run: npm run lint --prefix client
      - run: npm test --prefix server
      - run: npm test --prefix client
      - run: npm run build --prefix client
```

15.1 Production smoke test

- `GET /api/health` returns ok and db connected through the frontend domain.
- Home shows seeded projects; filters and Load more work; a project page opens and a shared link loads directly (SPA fallback works).
- Contact form submits; database record exists; notification email arrives.
- Admin: login works over HTTPS (cookie is Secure); create project with image upload; publish; it appears on Home; unpublish; it disappears; delete.
- View source shows correct title and meta on each route; `/sitemap.xml` and `/robots.txt` load; `/admin` is disallowed.
- Lighthouse mobile run on the production URL meets the targets in HLD section 11.

15.2 Rollback

Vercel and Render keep previous deployments: roll back the frontend by promoting the last good deployment; roll back the API by redeploying the previous commit. Database changes in v1 are additive, so no migrations need reversing.

16. Handover Requirements

- **README.md:** overview, prerequisites, setup, env variables, scripts, folder map, how to edit content files, how to add a project through the admin, deployment summary, troubleshooting.
- `.env.example` in both apps listing every variable with a comment.
- **Handover note (Markdown):** live URLs, admin login location (never the password in the repo), where placeholders must be replaced (brand, copy, prices, team, social links, OG image, logo), a list of manual accounts created, and known limitations.
- **Quality report:** test results, Lighthouse scores for Home, Pricing and one project page, and any deviations from this specification with reasons.

Final instruction to the build agent: build the complete product from scratch, phase by phase, verify each acceptance criterion, deploy, and report. Do not skip the design system: visual consistency and polish are the main differentiators of this project.